Toward a four-dimensional Yang–Mills construction

Notebook 27: an exact composite blocking step

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Abstract

Fix one primitive ordinary averaging ratio ℓ and take a finite macro ratio $R = \ell^s$. We construct the local chart and Gaussian of the actual composite constraint M_ℓ^s, using a separately chosen direct-R internal contour gauge. A selected interface bond has exactly one nonlinear consumer at every intermediate level. Its flat pivot coefficient is R^{1-d}, and elimination preserves all retained coordinates. A finite rectangular-patch argument gives coercivity of curvature, contour gauge and the composite constraint; the preceding classical action bounds then yield a retained Gaussian gap and exponentially decaying covariance uniformly in the old refinement. We derive the compact gauge-orbit normalization and the local Haar, delta, pivot, coupling and Gaussian factors of the one-shot constrained integral. Two minimum comparisons identify its physical centered branch. Its pre-gauge representative supplies a common-ball continuation of an inherited gauge-invariant insertion without changing integration variables or its cutoff. The construction is a finite small-field step. Uniform complex nonvanishing, supported inherited interaction bounds, amplitude control and the remaining sectors of the existence proof require their separate estimates.

The purpose of this construction is to make the factor R^{-1} of Notebook 25 available in an actual blocking integral. The primitive ratio remains fixed while $R = \ell^s$ is chosen large enough for the desired representative bound. The constraint is defined by its composition; no identity with a separately defined direct-R ordinary average is needed. Likewise, this one-shot integration is not identified with s already truncated primitive integrations. The gauge choice and every factor of the new integral are specified below. All old-depth uniformity refers to $M = \ell^k$, with ℓ, R fixed and compatible periodic volume.

1 Geometry of the actual composite ordinary constraint

Fix an odd primitive ratio $\ell \geq 3$, a positive integer s, and $R = \ell^s$. Current spacing is one. An earlier classical refinement $M = \ell^k$ does not enter any map in this section. Constants are uniform in M and compatible periodic volume once ℓ, s are fixed; no uniformity in $R \to \infty$ is asserted. Compatible source periods $2\ell^n$, $n \geq s$, suffice. Local paths are evaluated in their specified lifts, so the two opposite interfaces of a two-block periodic direction remain distinct. Translate the common block center to zero and identify the level-t lattice with $\ell^t\mathbb{Z}^d$.

The constraint constructed here is the actual composition of s ordinary primitive averages. It is not identified with a direct-R ordinary average or with s separately truncated quantum integrals. The direct-R contour variable used below is a separate gauge choice.

1.1 The source map and its exact covariance

For a centered primitive block and $x \in B_\ell(y)$, let $\mathcal{G}_\ell(y, x)$ contain the shortest coordinate-order paths from y to x. Every such path is internal to this block. Use the selected source group mean $\mathfrak{M}$ to define

$$P_\ell(V; y, x) = \mathfrak{M}(\{V(\Gamma) : \Gamma \in \mathcal{G}_\ell(y, x)\}). \tag{1}$$

For $c = (y, y + \ell e_\mu)$, put $x' = x + \ell e_\mu$ and let $R_c(V)$ be the straight center transporter. The ordinary map is

$$U_x(V) = P_\ell(V; y, x)V([x, x'])P_\ell(V; y + \ell e_\mu, x')^{-1},$$
$$\mathcal{M}_\ell(V)(c) = \exp\left\{\ell^{-d} \sum_{x \in B_\ell(y)} \log[U_x(V)R_c(V)^{-1}]\right\} R_c(V). \tag{2}$$

All logarithms use one common small branch. These are the ordinary average and contour definitions in [1], pp. 252–254, (0.3), (0.5)–(0.12).

The source axioms give local analyticity, the arithmetic differential at equal identity inputs, and endpoint covariance

$$\mathfrak{M}(\{gU_j h^{-1}\}) = g\mathfrak{M}(\{U_j\})h^{-1}. \tag{3}$$

For completeness, (0.10) of that paper supplies one admissible realization as the unique small solution P of $\sum_j \log(U_j P^{-1}) = 0$. The derivative with respect to $\log P$ at identity is minus the number of inputs times the identity, so the analytic implicit-function theorem gives uniqueness, analyticity and the arithmetic differential. Conjugating the defining logarithms proves (3) for this realization. The paper explicitly permits any mean satisfying its axioms; the proof here uses those properties for the selected mean and does not replace it by a different one.

With $V^u(b) = u(b_-)V(b)u(b_+)^{-1}$, it follows that

$$P_\ell(V^u; y, x) = u(y)P_\ell(V; y, x)u(x)^{-1},$$
$$\mathcal{M}_\ell(V^u) = \mathcal{M}_\ell(V)^{\,u|_{\text{new centers}}}. \tag{4}$$

Indeed every relative transporter inside the logarithm in (2) is conjugated at its source center. The identities hold on the common analytic domains and extend to complex fields there.

1.2 One primitive pivot has one nonlinear consumer

Select the unit interface bond

$$p_\ell(c) = [y + (\ell - 1)e_\mu/2,\ y + (\ell + 1)e_\mu/2]. \tag{5}$$

No internal contour reads this bond. A parallel string containing it must have direction μ, the same transverse block center, and the same pair of longitudinal endpoint blocks. Hence it belongs to the output equation c. The reference center string has the same property. This is a statement about the actual products and group means before differentiation:

$$\mathcal{M}_\ell(V)(c') \text{ is independent of } V(p_\ell(c)) \quad (c' \neq c). \tag{6}$$

The local lifted strings distinguish opposite periodic interfaces.

Let C_ℓ be the angle differential at identity, and let $t_y(B; x)$ be the arithmetic mean of the oriented sums of B on the root-to-x contours. Linearizing (2) gives

$$C_\ell B(c) = \ell^{-d} \sum_{x \in B_\ell(y)} \Big[t_y(B; x) + B([x, x + \ell e_\mu]) - t_{y+\ell e_\mu}(B; x + \ell e_\mu)\Big]. \tag{7}$$

The reference-string derivative cancels between the logarithms and the final factor. The root terms do not read the pivot. Exactly ℓ starting sites have its transverse coordinates, and each corresponding positive string contains it once. Thus its full column is

$$(C_\ell)_{c', p_\ell(c)} = \ell^{1-d}\delta_{c,c'} I_{\mathfrak{g}}. \tag{8}$$

This is an angle derivative, without division by the coarse length.

1.3 Exact nonlinear incidence through the composition

Define

$$\mathcal{M}_R^{\mathrm{comp}} = \mathcal{M}_\ell^s. \tag{9}$$

For a final bond $c = (y, y + Re_\mu)$ choose

$$p_R(c) = [y + (R-1)e_\mu/2,\ y + (R+1)e_\mu/2]. \tag{10}$$

Writing $h_t = \ell^t$, set

$$b_t(c) = [y + (R-h_t)e_\mu/2,\ y + (R+h_t)e_\mu/2], \qquad 0 \le t \le s. \tag{11}$$

Its endpoints are level-t sites because R/h_t is odd. Moreover $b_0 = p_R(c)$, $b_s = c$, and b_t is exactly the primitive middle-interface pivot of b_{t+1}.

Proposition 1.1. *Changing the input group coordinate at $p_R(c)$ changes at level t at most $b_t(c)$. Consequently every final constraint reads its own selected pivot and no other selected pivot. Each final output reads only unit bonds whose endpoints lie in its two endpoint R-blocks, using the prescribed local lift. If C_R denotes the composite angle differential at identity, then*

$$(C_R)_{c,p_R(c')} = R^{1-d}\delta_{c,c'} I_{\mathfrak{g}}. \tag{12}$$

Proof. The incidence assertion is induction along (11), applying (6) at spacing h_t. It is functional independence, so is valid for nonlinear variations, not only for the derivative. Distinct final bonds have distinct intermediate selected bonds.

For locality, a level-t output reads level-$(t-1)$ bonds in its two endpoint primitive blocks, including their connecting strings. The descendants centered at those intermediate sites tile the two centered blocks of side ℓ^t. Iterating gives the stated unit-bond dependency set. On short compatible tori this is read in the local cover, not inferred merely from periodic set containment.

Finally the chain rule along the single affected output multiplies s copies of $\ell^{1-d}I$. There is no intermediate row sum. The product is $R^{1-d}I$, proving (12). □

Composition of (4) gives

$$\mathcal{M}_R^{\mathrm{comp}}(V^u) = \mathcal{M}_R^{\mathrm{comp}}(V)^{\,u|_{R\text{-centers}}}. \tag{13}$$

With $D\phi(b) = \phi(b_+) - \phi(b_-)$, differentiate using $u = e^{-it\phi}$ to obtain

$$C_R D\phi(c) = \phi(c_+) - \phi(c_-). \tag{14}$$

The right side is an endpoint difference, or R times the physical coarse gradient.

1.4 Local analytic elimination and unchanged retained coordinates

For a small background $V = e^{ia}$ define

$$\begin{aligned} F_{R,V}(B;c) &= \frac{1}{i}\log\left[\mathcal{M}_R^{\mathrm{comp}}(e^{iB}V)(c)\mathcal{M}_R^{\mathrm{comp}}(V)(c)^{-1}\right], \\ A_{R,V} &= D_B F_{R,V}(0). \end{aligned} \tag{15}$$

Finite composition and the two-block dependency give a positive product domain in a, B, independent of volume and old M at fixed R. Put

$$\pi_c(V) = R^{d-1}(A_{R,V})_{c,p_R(c)}. \tag{16}$$

It is local analytic, $\pi_c(1) = I$, and on a common smaller domain $\|\pi_c - I\| \le 1/2$, $\|\pi_c^{-1}\| \le 2$. The right inverse inserts only at selected pivots:

$$(h_{R,V}w)(p_R(c)) = R^{d-1}\pi_c(V)^{-1}w(c), \qquad (h_{R,V}w)(b) = 0 \ (b \text{ nonpivot}), \qquad A_{R,V}h_{R,V} = I. \tag{17}$$

In particular its maximum row norm is at most $2R^{d-1}$. If $Q_{R,V} = A_{R,V}/R$ is the physical coarse-potential derivative, and its kernel is written as $QB(c) = \sum_b R^{-d}Q(c,b)B(b)$, that kernel's pivot coefficient is exactly $\pi_c(V)$. Thus the factors in (17) retain the source physical normalization.

Let E_f insert nonpivot values and zero at pivots, and abbreviate $A = A_{R,V}$, $h = h_{R,V}$. Define

$$K_{R,V} = E_f - hAE_f, \qquad AK_{R,V} = 0. \tag{18}$$

Restriction of $K_{R,V}z$ to nonpivots is z. Write $F_{R,V}(B) = AB + \mathcal{C}_V(B)$, with $\mathcal{C}_V(0) = D\mathcal{C}_V(0) = 0$, and solve

$$\widetilde{\mathcal{D}}_V(B) = \mathcal{C}_V(B - h\widetilde{\mathcal{D}}_V(B)). \tag{19}$$

For output c the right side reads only its own unknown $\widetilde{\mathcal{D}}_V(B;c)$: other unknowns insert at pivots that the full nonlinear c equation does not read. The implicit derivative at zero is identity. A finite-dimensional analytic implicit theorem, or contraction after imposing $\|D\mathcal{C}_V h\| < 1/2$, therefore gives a unique local analytic solution. There are finitely many oriented neighborhood types, so one radius works in all volumes. Iteration does not spread its spatial dependency.

The exact retained chart is

$$\begin{aligned} \Phi_{R,V}(z) &= K_{R,V}z - h\widetilde{\mathcal{D}}_V(K_{R,V}z), \\ F_{R,V}(\Phi_{R,V}(z)) &= 0, \qquad \Phi_{R,V}(z)(b) = z(b) \quad (b \text{ nonpivot}). \end{aligned} \tag{20}$$

It is the unique small solution with its prescribed retained values. Direct differentiation gives

$$\begin{aligned} D\Phi_{R,V}(0) &= K_{R,V}, \\ D^2\Phi_{R,V}(0)[v,w] &= -hD^2F_{R,V}(0)[K_{R,V}v, K_{R,V}w]. \end{aligned} \tag{21}$$

The affine equation $F_{R,V}(B) = w$ is equally local:

$$B = K_{R,V}z + hw - h\mathcal{C}_V(B). \tag{22}$$

It has a common positive small product domain in a, w, z and still retains z exactly. Local finite dimension, Cauchy estimates and bounded neighborhood incidence give finite weighted first-derivative row and column bounds and finite contracted second-derivative bounds. Their constants may depend on R, but not on M or volume.

After coupling insertion the relative field is $\Phi_{R,V}(gz)$, whose retained values equal gz. It is not $g\Phi_{R,V}(z)$. Additional pivot, contour or reconstructed field indicators remain at their exact nonlinear arguments.

1.5 An independent direct-R contour gauge

Use the same selected source mean on root-to-site coordinate-order paths inside one centered R-block:

$$P_R(V;y,x) = \mathfrak{M}(\{V(\Gamma) : \Gamma \in \mathcal{G}_R(y,x)\}). \tag{23}$$

This does not redefine $\mathcal{M}_R^{\text{comp}}$. Endpoint covariance gives

$$P_R(V^u;y,x) = u(y)P_R(V;y,x)u(x)^{-1}. \tag{24}$$

Every path is internal to its block; every selected pivot crosses a block interface. Thus P_R is exactly independent of all selected pivot group variables at every allowed background.

Define

$$T_{R,V}B(y,x) = D_B \frac{1}{i}\log[P_R(e^{iB}V;y,x)P_R(V;y,x)^{-1}]\Big|_{B=0}. \tag{25}$$

The pivot support of $h_{R,V}$ then gives

$$T_{R,V}h_{R,V} = 0, \qquad T_{R,V}K_{R,V} = T_{R,V}E_f. \tag{26}$$

More strongly,

$$P_R(e^{i\Phi_{R,V}(z)}V;y,x) = P_R(e^{iE_f z}V;y,x). \tag{27}$$

The contour penalty therefore reads exactly the unchanged retained variables, although its coefficients still depend on V.

For normalized trace τ put

$$\mathcal{G}_R(V) = \sum_y \sum_{\substack{x\in B_R(y)\\ x\neq y}} \left[1 - \tfrac{1}{2}\tau(P_R(V;y,x) + P_R(V;y,x)^{-1})\right]. \tag{28}$$

It is nonnegative on real unitary fields. At a background with every $P_R = I$ its first derivative vanishes and its pulled-back Hessian, in counting/trace coordinates, is

$$D_z^2\mathcal{G}_R(e^{i\Phi_{R,V}(z)}V)\big|_{z=0} = K_{R,V}^T T_{R,V}^T T_{R,V}K_{R,V}. \tag{29}$$

The second derivative of the chart contributes nothing because it is pivot supported and the gauge action is pivot independent. At identity, (24) also gives

$$T_{R,1}D\phi(y,x) = \phi(x) - \phi(y). \tag{30}$$

1.6 Clearing this gauge and the domain boundary

Set $u_V(y) = I$ at all final centers and $u_V(x) = P_R(V;y,x)$ elsewhere in its block. Then, on the common domain,

$$P_R(V^{u_V};y,x) = I, \qquad \mathcal{M}_R^{\mathrm{comp}}(V^{u_V}) = \mathcal{M}_R^{\mathrm{comp}}(V). \tag{31}$$

This is a geometric gauge operation, not by itself a minimum or stationarity argument.

There are at most $d!$ contour orderings and every path has length at most $d(R-1)/2$. In a fixed matrix norm, $\|\prod_b e^{ia_b} - I\| \le \exp(\sum_b\|a_b\|) - 1$. The selected local mean has a fixed Lipschitz constant near equal identity inputs, with the number of inputs bounded by $d!$. Consequently, when $R\|a\|_\infty$ is sufficiently small,

$$\|\log u_V/i\|_\infty \le CR\|a\|_\infty. \tag{32}$$

The constant here depends on dimension, group and the selected mean, not on R. For a pre-gauge representative $a_p = O(w/R)$ this gives $u = O(w)$. It does not give a rooted link-angle bound $O(w/R)$: endpoint gauge differences may have size $O(w)$. Nor does (32) make the local elimination radius uniform in R.

The composite constraint, its actual local chart, and the independent contour gauge are thus established. Gaussian coercivity and locality, the physical minimum comparisons, the exact compact gauge insertion and all cutoff normalizations are separate modules. None of these geometric identities supplies a repeated quantum RG induction.

2 The retained Gaussian for the composite macrostep

Fix an odd primitive ratio $\ell \geq 3$, a finite macro ratio $R = \ell^s$, $s \geq 1$, and an arbitrary preceding classical depth $M = \ell^k$, $k \geq 0$. Current lattice spacing is one. All constants in this section may depend on the fixed ℓ, s, but are independent of M and compatible periodic volume. We use the actual composite ordinary constraint $F_R = \mathcal{M}_\ell^s$ and an independently chosen direct-R internal contour gauge. No equality with a direct-R ordinary average is used.

2.1 Established classical inputs and the actual flat constraint

Let $S_M(a)$ be the homogeneous ordinary classical minimum action at current datum e^{ia}. Its flat Hessian q_M has the previously established variational description

$$\langle B, q_M B\rangle = \inf_{Q_M^{\mathrm{st}} A = B} \|d_{1/M} A\|_{1/M}^2, \qquad \|d_1 B\|_2^2 \leq \langle B, q_M B\rangle \leq C_d \|B\|_2^2, \quad C_d = 2^{d+6}(d-1). \tag{33}$$

The case $M = 1$ is included. In current counting norms, q_M has a uniform weighted Schur bound at some positive exponent κ_q. On a common independent-background product ball, the earlier classical construction also gives

$$DS_M(0) = 0, \qquad \|DS_M(a)\|_\infty \leq C\|a\|_\infty, \qquad \|D^2 S_M(a) - q_M\|_\kappa \leq C_\kappa \|a\|_\infty \tag{34}$$

at a sufficiently small positive exponent. These are inputs from the homogeneous classical action; their constants do not depend on the next chosen macro ratio.

Define the composite *angle* differential

$$C_R B = \frac{d}{dt}\frac{1}{i}\log F_R(e^{itB})\Big|_{t=0}. \tag{35}$$

It is not divided by R. The exact composite geometry supplies the following facts. A row for the final bond $c = (y, y + Re_\mu)$ reads only unit bonds in the union of its two endpoint R-blocks. Endpoint gauge covariance gives

$$C_R D\phi(c) = \phi(y + Re_\mu) - \phi(y), \qquad D\phi(b) = \phi(b_+) - \phi(b_-). \tag{36}$$

Choose the unit centerline interface pivot

$$p(c) = [y + (R-1)e_\mu/2,\ y + (R+1)e_\mu/2]. \tag{37}$$

At level t its unique affected intermediate bond has endpoints $y + (R - \ell^t)e_\mu/2$ and $y + (R + \ell^t)e_\mu/2$. Thus the primitive pivot coefficient ℓ^{1-d} multiplies along one chain and yields

$$(C_R)_{c,p(c')} = R^{1-d}\delta_{cc'} I. \tag{38}$$

These statements hold in the periodic covering geometry with the specified lifted paths and distinct opposite interfaces.

Let E_f insert nonpivot coordinates and zero at pivots. Define

$$(h_R w)(p(c)) = R^{d-1} w(c), \qquad K_R = (I - h_R C_R) E_f. \tag{39}$$

Then $C_R h_R = I$, $C_R K_R = 0$, K_R retains its input coordinates, and $\operatorname{ran} K_R = \ker C_R$. Both maps have fixed R-range. One conservative finite norm bound is

$$r_\ell = 1 + \frac{2d(\ell-1)}{\ell}, \qquad c_R = \left[\ell r_\ell \sqrt{2d+2}\right]^s, \qquad \|C_R\| \leq c_R, \qquad \|K_R\|^2 \leq k_R^2 := 1 + R^{2d-2} c_R^2. \tag{40}$$

Indeed the primitive counting norm is bounded by $\ell r_\ell \sqrt{2d+2}$ and these angle spaces compose. The retained and pivot parts of $K_R z$ have disjoint coordinate supports, which proves the last bound.

2.2 The independent contour gauge and a local nullspace argument

Use the actual direct-R internal averaged contours solely as gauge fixing, with flat derivative

$$(T_R B)(y,x) = \mathrm{av}_{\Gamma: y\to x} \sum_{b\in\Gamma} \mathrm{or}(b) B_b. \tag{41}$$

Every contour stays in its own centered R-block. Consequently

$$T_R D\phi(y,x) = \phi(x) - \phi(y), \qquad T_R h_R = 0. \tag{42}$$

For normalized trace τ, the absolute real gauge action $G_R(V) = \sum_{y,x}[1 - \mathrm{Re}\,\tau(P_R(V;y,x))]$ has zero constant and linear terms at $V = 1$, and flat Hessian $T_R^T T_R$. This follows from $P_R(e^{itB}) = I + itT_R B + O(t^2)$ and unitarity, or from its local logarithm. In the complex domain use $1-\tau(P+P^{-1})/2$. The elementary row/column estimate gives

$$\|T_R\|^2 \le t_R^2 := d(R-1)(R^d-1). \tag{43}$$

The two factors bound contour length and the number of target rows through an internal bond. This is the actual averaged contour penalty, not an assumed tree-coordinate penalty.

Let Ω be the rectangular cube consisting of the 3^d centered R-blocks indexed by $\{-1,0,1\}^d$. Include all positive unit bonds with both endpoints in Ω and all unit plaquettes with vertices there. Let $T_{R,\Omega}$ contain all internal contour rows in these blocks, and let $C_{R,\Omega}$ contain the neighboring-center rows whose two centers belong to this cube. The two-block locality places every one of these rows inside the chosen bond set. Set

$$\mathcal{L}_{R,\Omega} = d_\Omega^T d_\Omega + T_{R,\Omega}^T T_{R,\Omega} + C_{R,\Omega}^T C_{R,\Omega}, \qquad \sigma_R = \lambda_{\min}(\mathcal{L}_{R,\Omega}). \tag{44}$$

Lemma 2.1 (Local positivity)**.** *The finite scalar matrix in* (44) *has* $\sigma_R > 0$.

Proof. If its quadratic form vanishes, $d_\Omega B = 0$. The rectangular grid is simply connected: its plaquette relations imply path independence and hence $B = D\phi$ on every bond of Ω. Equation (42) makes ϕ constant on each R-block. Equation (36) makes the constants on neighboring blocks equal. Its local use is justified by extending ϕ arbitrarily outside Ω and then using two-block locality of the row. The coarse block graph is connected, so ϕ is constant on all of Ω and $B = 0$. The matrix therefore has trivial kernel and is positive definite. □

This explicitly excludes the gauge and harmonic obstructions. Mere finiteness of a local matrix would not have supplied the required positivity.

Translate the patch over all R-block centers. Every bond occurs at least once, whereas every plaquette, contour row and constraint row occurs at most 3^d times. Thus

$$\sigma_R \|B\|_2^2 \le 3^d(\|d_1 B\|_2^2 + \|T_R B\|_2^2 + \|C_R B\|_2^2). \tag{45}$$

On a compatible torus, pull the periodic field back to the ordinary covering cube and retain repeated occurrences. Summing the same local inequalities has exactly the same multiplicity bound; an embedded nonwrapping patch is not required. The distinct-interface assumptions of the composite chart remain in force.

Combining (33) and (45), for $B \in \ker C_R$ we obtain

$$\langle B, q_M B\rangle + \|T_R B\|_2^2 \ge \gamma_R \|B\|_2^2, \qquad \gamma_R = \sigma_R/3^d > 0. \tag{46}$$

The upper bound on arbitrary B is $(C_d + t_R^2)\|B\|_2^2$. None of these constants depends on M.

2.3 Precision, covariance and weighted inverse

At flat background both scalar gradients vanish. Therefore the second derivative of the nonlinear constraint chart contributes no gradient-times-chart-curvature term. The exact retained flat precision is

$$A_{M,R,0} = K_R^T(q_M + T_R^T T_R)K_R. \tag{47}$$

Since retained restriction of $K_R z$ is z,

$$\gamma_R I \le A_{M,R,0} \le B_R I, \qquad B_R = (C_d + t_R^2)k_R^2, \qquad B_R^{-1} I \le A_{M,R,0}^{-1} \le \gamma_R^{-1} I. \tag{48}$$

There are $d(R^d - 1)\dim \mathfrak{g}$ retained real coordinates per R-block. Their normalized Gaussian has density

$$(2\pi)^{-n/2} \det(A_{M,R,0})^{1/2} \exp[-\tfrac{1}{2} z^T A_{M,R,0} z]\, dz. \tag{49}$$

Its covariance is the counting inverse $A_{M,R,0}^{-1}$. There is no original-fine volume factor M^d in these current coordinates. Haar and nonlinear Jacobian terms remain in the interaction.

For a block matrix, write $\|A\|_{\mathrm{Sch},\alpha}$ for the maximum of its exponentially weighted row and column sums of block operator norms. Regrouping the current q_M bound into R-blocks gives, for $\alpha/R \le \kappa_q$,

$$\|q_M\|_{\mathrm{Sch},\alpha;R} \le R^d e^{\alpha d} K_q(\alpha/R). \tag{50}$$

The fixed-range maps K_R, T_R and weighted matrix multiplication then give some $\alpha_R > 0$ and $A_R^* < \infty$ such that

$$\|A_{M,R,0}\|_{\mathrm{Sch},\alpha_R} \le A_R^* \tag{51}$$

uniformly in M and compatible volume.

Here is an explicit inverse argument. Fix an R-block j and put $W_{ii} = e^{td(i,j)} I$. Periodic distance is used on a torus. For $0 < t \le \alpha_R/2$, both the row and column Schur bounds of $WAW^{-1} - A$ are at most

$$\sup_i \sum_k \|A_{ik}\|(e^{td(i,k)} - 1) \le \frac{2tA_R^*}{e\alpha_R}. \tag{52}$$

Indeed $e^{tx} - 1 \le txe^{tx}$ and $\sup_{x \ge 0} xe^{-\alpha_R x/2} = 2/(e\alpha_R)$. Choose

$$t_R^{\mathrm{loc}} = \min\left\{ \frac{\alpha_R}{2}, \frac{e\alpha_R \gamma_R}{4A_R^*} \right\}. \tag{53}$$

The perturbation has Hilbert norm at most $\gamma_R/2$. The resolvent identity and (48) give $\|(WAW^{-1})^{-1}\| \le 2/\gamma_R$, whence

$$\|(A_{M,R,0}^{-1})_{ij}\| \le \frac{2}{\gamma_R} e^{-t_R^{\mathrm{loc}} d(i,j)}. \tag{54}$$

On the infinite lattice, truncated distance weights give the same estimates and may be removed in each matrix entry. At $\kappa_R = t_R^{\mathrm{loc}}/2$ this implies

$$\|A_{M,R,0}^{-1}\|_{\mathrm{Sch},\kappa_R} \le \frac{2}{\gamma_R} \sum_{z \in \mathbb{Z}^d} e^{-t_R^{\mathrm{loc}}|z|_1/2}. \tag{55}$$

Periodic sums are bounded by this same lattice sum by shortest representatives. The proof uses both the real gap and the already-established decay of q_M; a spectral sandwich alone would not imply locality.

2.4 The independent-background extension

Let $\Phi_a(z)$ be the actual finite-range analytic pivot chart for the composite relative constraint at $V = e^{ia}$, retaining z exactly. Define

$$a_V(z) = \frac{1}{i}\log[e^{i\Phi_a(z)}e^{ia}], \qquad A_{M,R}(a) = D_z^2\{S_M(a_V(z)) + G_R(e^{i\Phi_a(z)}e^{ia})\}\Big|_{z=0}. \tag{56}$$

The action contribution is exactly

$$(Da_V)^T D^2 S_M(a) Da_V + \langle DS_M(a), D^2 a_V\rangle. \tag{57}$$

Thus both constraint-straightening curvature and ordinary-log curvature are retained. The gauge contribution is its full absolute Hessian, including its residual away from $P_R = I$.

The local chart derivatives have fixed-R row/column bounds on a smaller product ball. Contracting their local second derivatives against the supremum bound for $DS_M(a)$, and using (34), proves at a smaller fixed exponent

$$\|A_{M,R}(a) - A_{M,R,0}\|_{\mathrm{Sch},\kappa_R} \le C_{R,\kappa_R}\|a\|_\infty. \tag{58}$$

Every contraction is a finite local incidence sum; no extensive gradient norm or old-M coordinate count occurs. A sufficiently small background ball gives complex accretivity and a holomorphic weighted inverse by its Neumann series. For example, one may require both $C_{R,\kappa_R}\|a\|_\infty < \gamma_R/2$ and $\|A_{M,R,0}^{-1}\|_{\mathrm{Sch},\kappa_R} C_{R,\kappa_R}\|a\|_\infty < 1/2$.

These are uniform-old-depth Gaussian statements for the chosen composite constraint and independent contour gauge. Physical centering requires the separate stationary-branch comparison; the exact Haar, delta, gauge-insertion and cutoff normalization requires the separate integral argument. Strict background carrier coefficients require the classical support composition, not merely (58). This section proves neither a quantum induction nor a uniform estimate as R tends to infinity.

3 The exact composite integral and its pre-gauge continuation

Fix the primitive odd ratio ℓ and a macro ratio $R = \ell^s$. The preceding refinement is $M = \ell^k$. All statements concern finite compatible periodic regulators. Constants may depend on the fixed R, ℓ, but are uniform in M and volume. The constraint is the actual composite

$$\mathcal{F}_R = \mathcal{M}_\ell^s.$$

It is not replaced by a direct-R ordinary average. The direct-R internal contour variable P_R is used independently for gauge fixing. Nor is the one-shot integral below identified with s separately gauge-fixed and truncated quantum integrations.

3.1 A gauge insertion with its exact normalization

Use normalized Haar measure on the compact matrix group G, and the convention $V^u(b) = u(b_-)V(b)u(b_+)^{-1}$. The previously proved local geometry gives

$$\begin{aligned}\mathcal{F}_R(V^u) &= \mathcal{F}_R(V)^{u|_{\text{final centers}}},\\ P_R(V^u; y, x) &= u(y)P_R(V; y, x)u(x)^{-1}.\end{aligned} \tag{59}$$

Every contour defining P_R stays inside its R-block. Put

$$v(h) = 1 - \mathrm{Re}\,\tau h, \qquad \mathcal{G}_R(V) = \sum_y \sum_{\substack{x\in B_R(y)\\ x\neq y}} v(P_R(V; y, x)). \tag{60}$$

The analytic continuation of $v(h)$ is $1 - \frac{1}{2}\tau(h + h^{-1})$. This is the absolute contour penalty. It is not replaced by a relative penalty away from a gauge-zero background.

A local logarithmic formula does not by itself allow integration over every compact gauge transformation. One may work on a gauge-invariant domain where the contour formula is defined on complete orbits, or make the following explicit measurable completion. For a path tuple (U_j) from y to x, retain its specified reference T. On a fixed small conjugation-invariant neighborhood of the identity for every $U_j T^{-1}$, put

$$P(U_1, \dots, U_n) = \Phi(U_1 T^{-1}, \dots, U_n T^{-1})T,$$

where Φ is the *actual selected local source mean* on the reference-normalized tuple, with its original contour multiplicities, ordering and reference. Right endpoint covariance proves this representation on the original domain, and simultaneous conjugation covariance follows from the same source covariance. On the complementary set put $P = T$. Thus the completion is measurable, group valued and endpoint covariant everywhere, and equals the chosen analytic mean near the identity. No different mean or additional mean equation is substituted.

Similarly, for each primitive ordinary average, use its original logarithmic formula when all its reference-normalized transported strings lie in the chosen small logarithm neighborhood, and its reference string otherwise. The condition is conjugation invariant. This gives a measurable gauge-covariant completion of $\mathcal{M}_\ell$, hence of its composition $\mathcal{F}_R$. The values outside the analytic domain are part of this proposed full blocking map; equality with a different large-field completion is not claimed. The following local formulas agree wherever the original analytic maps agree.

Let $\mathcal{H}_R$ be the group of current-site gauges equal to one at final block centers, with normalized independent Haar measure at the n_s other sites. For $g > 0$ define

$$c(g) = \int_G e^{-v(h)/g^2}\, dh > 0. \tag{61}$$

Then

$$\int_{\mathcal{H}_R} e^{-\mathcal{G}_R(V^u)/g^2}\, du = c(g)^{n_s}. \tag{62}$$

Indeed, by (59), the factor at (y, x) is $e^{-v(P_R(V;y,x)u(x)^{-1})/g^2}$, whose integral is $c(g)$ by Haar invariance, independently of V. The site integrals factor. There is no omitted gauge determinant: this is direct integration over compact Haar variables with the displayed normalization.

For an integrable gauge-invariant current density ρ, and any bounded test function f of the final configuration,

$$\int dV\, \rho(V) f(\mathcal{F}_R(V)) = c(g)^{-n_s} \int dV\, \rho(V) f(\mathcal{F}_R(V)) e^{-\mathcal{G}_R(V)/g^2}. \tag{63}$$

Insert (62), use Fubini, then change V by the fixed gauge u; link Haar, ρ and $\mathcal{F}_R$ are unchanged. This is an identity of finite pushforward measures, requiring no globally regular delta density. It also holds for a finite gauge-invariant current measure, and for a gauge-invariant orbit-domain restriction.

A noninvariant small-field window is introduced *after* this insertion. Splitting the gauged integral into χ_{sf} and $1 - \chi_{\mathrm{sf}}$ is exact for any measurable $0 \le \chi_{\mathrm{sf}} \le 1$ on the fiber. If ρ is positive both pieces are positive on real fields. Neither piece is asserted to satisfy (63) without the gauge factor. The complement requires a separate large-field estimate.

3.2 Identification of the physical centered branch

Let $U_f(W)$ be the proved full ordinary minimizer at total refinement MR, with final spacing R in current units, and set

$$V_p(W) = \mathcal{M}_\ell^k(U_f(W)), \qquad \mathcal{F}_R(V_p(W)) = W. \tag{64}$$

Let $U_c(V)$ be the preceding-depth ordinary minimizer and $S_M(V) = S(U_c(V))$. Choose a common current comparison radius δ_R and then a final datum radius b_R so that $U_f(W)$ belongs to the current minimum class, every $U_c(V)$ with $\|\log V/i\|_\infty < \delta_R$ belongs to the full final-spacing-R minimum class, and the partial average and gauges below stay within all common chart and invariance domains. These are finitely many strict smallness conditions with R-dependent constants, uniform in M and volume. In dimension four the coupling-free Wilson action has no additional scale factor; the potential norms acquire only fixed powers of R.

The two actual comparisons are

$$S_M(V_p) \le S(U_f) \le S(U_c(V_p)) = S_M(V_p). \tag{65}$$

For the second inequality exact composition makes $U_c(V_p)$ a full feasible competitor. For any other current V in the comparison ball with $\mathcal{F}_R(V) = W$, the same argument gives

$$S_M(V) \ge S(U_f) = S_M(V_p). \tag{66}$$

This is a common-domain minimum comparison, not a dynamic-programming assertion for arbitrary backgrounds.

Define the root-fixed current gauge and background by

$$u(y) = 1, \qquad u(x) = P_R(V_p; y, x), \qquad V_*(W) = V_p(W)^u. \tag{67}$$

Then $\mathcal{F}_R(V_*) = W$ and $P_R(V_*; y, x) = 1$ exactly. At fixed R this is local analytic with $\|\log u/i\|_\infty \le C_R\|\log V_p/i\|_\infty$. The already proved coarse-action gauge invariance, with its smooth fine point-value extension, implies $S_M(V_*) = S_M(V_p)$ after shrinking the common radii. Since $\mathcal{G}_R \ge 0$ and $\mathcal{G}_R(V_*) = 0$ on real fields, V_* is a local constrained minimum of $S_M + \mathcal{G}_R$.

The preceding Gaussian module supplies its uniformly positive flat retained precision. The local analytic chart and $D^2 S_M = q_M + O(\|\log V\|_\infty)$ show that the pulled-back Hessian on a common convex retained ball is $A_{M,0} + O_R(\|w\|_\infty + \|z\|_\infty)$. It remains positive after a fixed shrink. Therefore the centered real critical point is unique in that common neighborhood; gauge zero alone would not prove uniqueness. Constraint, gauge-zero and first-derivative identities continue holomorphically to complex W. No complex minimum is asserted.

The supported pre-gauge theorem gives the stronger amplitude bound

$$a_p = \log V_p/i, \qquad \|a_p\|_\infty \le C_a\|w\|_\infty/R. \tag{68}$$

Its supported-family constant is fixed at primitive ℓ independently of s, k. There is no corresponding $1/R$ contraction claim for the rooted potential $\log V_*/i$.

3.3 The local density, Haar factors and pivot determinants

Use Hermitian Lie coordinates orthonormal for τ, of dimension $q = \dim \mathfrak{g}$. Let n_e and n_c count positive current and final macro bonds, and put $n_z = q(n_e - n_c)$. In exponential coordinates write normalized Haar as

$$d(e^{iX}) = h_0 j(X)\, dX, \qquad h_0 > 0, \quad j(0) = 1. \tag{69}$$

Consequently $\delta_G(e^{iF}) = h_0^{-1}\delta_{\mathbb{R}^q}(F)$, with the coordinate measure understood: $j(F)$ is evaluated at $F = 0$.

Fix a real small W and $V = V_*(W)$. The composite relative constraint and its pivot derivative are

$$F_V(B) = \log[\mathcal{F}_R(e^{iB}V)\mathcal{F}_R(V)^{-1}]/i, \qquad A_V = DF_V(0), \qquad A_{c,p(c)} = R^{1-d}\pi_c(V). \tag{70}$$

Let ψ_V be the ambient straightening, K_V the linear retained insertion, and $\Phi_V(z) = \psi_V(K_V z)$. Each retained component of Φ_V equals z exactly. Order ambient variables as retained coordinates followed by pivots. The derivative of ψ_V is block triangular; its pivot diagonal blocks satisfy

$$D_{p(c)} F_V(\psi_V(B); c)\, D_{B(p(c))} \psi_V(B)(p(c)) = A_{c,p(c)}.$$

Thus

$$\det D\psi_V(B) = \prod_c \frac{\det A_{c,p(c)}}{\det D_{p(c)} F_V(\psi_V(B); c)}. \tag{71}$$

Let $j_c^{\mathrm{str}}(V; B)$ be the analytic logarithm of the cth ratio, zero at $B = 0$, and put

$$\mathcal{D}_R(V) = \prod_c (\det A_{c,p(c)})^{-1}. \tag{72}$$

The determinants have their positive flat orientation on the real small chart.

Choose the small-field sector to be the component of this relative chart with retained coordinates in a specified product window, retaining any additional indicator explicitly. Changing the square ambient variables and then solving $A_V B = 0$ gives its exact density

$$c(g)^{-n_s} h_0^{n_e - n_c} \mathcal{D}_R(V) \int dz \prod_b j(\Phi_V(z)(b))\, e^{\sum_c j_c^{\mathrm{str}}(V; K_V z)} \rho(e^{i\Phi_V(z)} V) e^{-\mathcal{G}_R(e^{i\Phi_V(z)} V)/g^2}. \tag{73}$$

The integration domain and every extra indicator are those of the chosen sector. This uses the square Jacobian (71), not a determinant of the rectangular derivative $D\Phi_V$. Equivalently the unrearranged pivot density is the product of inverse $D_{p(c)} F_V(\Phi_V(z); c)$ determinants.

Here the globally gauge-invariant ρ in (63) is an input density or measure. Only on the common small chart do we use a representation

$$\rho(U) = \exp[-S_M(U)/g^2 + E(U)],$$

with any field-independent factor tracked separately. The local classical branch S_M has not been promoted to a global action on all configurations.

Put $z = g\xi$ for real $g > 0$, and let $A = A_M(V)$ be the exact retained Hessian of $S_M + \mathcal{G}_R$. Define

$$\begin{aligned} \mathcal{R}_3(V, z) &= S_M(e^{i\Phi_V(z)} V) + \mathcal{G}_R(e^{i\Phi_V(z)} V) - S_M(V) - \tfrac{1}{2}\langle z, Az\rangle, \\ I(V, g, \xi) &= -g^{-2} \mathcal{R}_3(V, g\xi) + \sum_c j_c^{\mathrm{str}}(V; K_V g\xi) + \sum_b \log j(\Phi_V(g\xi)(b)) \\ &\quad + E(e^{i\Phi_V(g\xi)} V) - E(V). \end{aligned} \tag{74}$$

There is no linear term because the background is the physical centered branch. Formula (73) becomes

$$\begin{aligned} c(g)^{-n_s} h_0^{n_e - n_c} \mathcal{D}_R(V)\, g^{n_z} (2\pi)^{n_z/2} \det(A)^{-1/2} e^{-S_M(V)/g^2 + E(V)} \\ \times \mathbb{E}_{A^{-1}} \left[\chi_{\mathrm{window}}(\xi) e^{I(V, g, \xi)} \right]. \end{aligned} \tag{75}$$

The Gaussian convention is the retained Euclidean density $\det(A)^{1/2} (2\pi)^{-n_z/2} e^{-\langle \xi, A\xi\rangle/2}$. The covariance is that of this exact classical action-plus-gauge Hessian; Haar remains in (74). The factors $c(g)^{-n_s}$, $h_0^{n_e - n_c}$ and g^{n_z} are vacuum normalizations. $\mathcal{D}_R(V)$ and $\det A$ generally depend on the background and are not vacuum constants.

3.4 The literal retained cutoff

For the original retained link/color balls set

$$\chi_{\text{window}}(\xi) = \prod_{b \text{ retained}} \mathbf{1}_{\{|\xi(b)| \le T\}}.$$

Since $\Phi_V(g\xi)(b) = g\xi(b)$ at every retained bond, this is exactly the relative-coordinate window of the chosen sector. It is not a cutoff on $K_V\xi$, or a Euclidean ball on a whole macro fiber. Pivot, contour, plaquette or reconstructed-field indicators, if included, remain at their exact $\Phi_V(g\xi)$ pullbacks. The macro sector is not identified with an intersection of the primitive windows of s separate integrations.

At fixed R, the normalized pivot logarithms are local $O_R(|g|T)$ terms. Inversion invariance gives $\log j(X) = O(|X|^2)$, so Haar logarithms are local $O_R(g^2T^2)$ terms. The gauge third-and-higher remainder is local $O_R(|g|T^3)$. The internal contours read no pivot, and therefore its full exact factor satisfies

$$\mathcal{G}_R(e^{i\Phi_V(z)}V) = \mathcal{G}_R(e^{iE_{\text{free}}z}V). \tag{76}$$

On an independent background product ball its constant and linear terms must also be separated away from the physical gauge-zero branch. All these estimates have finite local supports and finite anchor multiplicities at fixed R, without a sum over the old refinement M. The classical action and inherited interaction require their own support estimates.

3.5 Pre-gauge continuation of the full inherited insertion

With $V_* = V_p^u$, set, link by link,

$$B_p(w,\xi)(b) = \mathrm{Ad}_{u(w;b_-)^{-1}}\, \Phi_{V_*(w)}(g\xi)(b). \tag{77}$$

Direct multiplication proves

$$\left(e^{i\Phi_{V_*}(g\xi)}V_*\right)^{u^{-1}} = e^{iB_p(w,\xi)}V_p(w). \tag{78}$$

The coupling remains inside the nonlinear chart: $\Phi_V(g\xi)$ is not replaced by $g\Phi_V(\xi)$. The gauge depends on w, not on ξ. Its real adjoint is a color isometry; on the small complex gauge ball it has a fixed R-dependent bound.

Let E be a holomorphic total inherited functional on a current potential product ball of radius b, with its local gauge invariance on the original common domain. Individual activity coefficients need not be gauge invariant. On a sufficiently small common overlap, both arguments of (78), and a small connecting gauge path, are in this domain. Their E values therefore agree, first on real fields and then by holomorphic uniqueness. If only e^E is supplied, a compatible nonvanishing holomorphic logarithm branch must be supplied before making the corresponding statement about E.

On a fixed target b-ball below all classical domains, (68) and the local chart give

$$\left\| \frac{1}{i} \log[e^{iB_p(w,\xi)}V_p(w)] \right\|_\infty \le C_a b/R + C_R|g|T. \tag{79}$$

Choose $R \ge 4C_a$ and $C_R|g|T \le b/4$. The pre-gauge argument then stays inside the old b-ball uniformly in M and volume. Its inherited functional is defined and holomorphic on the whole target ball, even where the rooted argument falls outside the original domain of E. It is a continuation of the common-domain germ, not an assertion that E was already defined at that rooted argument. Replacing $g\xi$ by $\lambda g\xi$ gives a complex λ-disc of radius at least $cb/(C_R|g|T)$ within the strict chart margins.

The *entire* inherited contribution in (75) is continued: replace its outside $E(V_*)$ by $E(V_p)$, and replace the inside increment by

$$E(e^{iB_p(w,\xi)}V_p(w)) - E(V_p(w)).$$

Leaving $E(V_*)$ outside would be undefined on part of the enlarged domain. There is no change of integration variable ξ. The Gaussian, its determinant, the pivot and Haar factors, the absolute gauge penalty and the retained cutoff remain exactly those of (75); no adjoint Jacobian or transformed window is introduced.

3.6 What holomorphy does and does not imply

At fixed finite regulator and fixed real retained window, the continued integrand is holomorphic in w. The proved complex precision neighborhood supplies a consistent analytic determinant square root. Every compact subdomain has an integrable uniform majorant, so the fixed-window integral is holomorphic by differentiation under the integral. Equality on the nonempty overlap identifies it as the continuation of the original small-field germ.

For real data and positive physical input, the exact integrand is positive and the retained window has positive measure. The integral is then strictly positive. This does not imply nonvanishing on a volume-independent complex ball: an integral of pointwise nonzero analytic factors can vanish. A finite-volume logarithm germ is available near a positive point; a uniform complex logarithm needs the full cluster hypotheses and budgets.

Similarly, pointwise λ-Cauchy applied to a total E can give only an extensive estimate. An anchored interaction bound needs a strict joint (w, ξ) family for the pre-gauge argument, with remote background supports and an explicit local fluctuation collar. That separate support module is not inferred from (79). Purely classical coefficients have independent fixed classical domains; the quantum radius may be chosen below them once, rather than reused as their outer analytic radius at every step.

The present module proves the finite-regulator normalization and local density identities, identifies the physical centered branch, preserves the exact window, and supplies the pointwise pre-gauge extension. It does not establish a stable quantum iteration, uniform nonvanishing without the cluster bounds, large-field control, mixed-history matching, coupling flow or a continuum measure.

4 Place in the construction

The longer physical blocking step is now defined, including the source chart, its centered Gaussian, its gauge insertion and its small-field cutoff. The pre-gauge representative of Notebook 25 belongs to this actual composite constraint. Choosing R after its primitive-base constants have been fixed can therefore provide a strict amplitude margin for an inherited field on a fixed analytic ball. This is the role of the new macrostep.

The next analytic task is to control the full inherited insertion as strict joint activities in the coarse background and retained fluctuations. A local-core expansion can keep its nearby field dependence unexpanded and spend Taylor estimates only on its distant tail. The precision and purely classical interactions can be expanded directly in the coarse background before Gaussian integration, using their existing finite classical support reserves. Both operations require their explicit norm budgets; the pointwise continuation proved here is not a substitute.

Even a common analytic domain and a convergent one-step logarithm would leave a noncontracting activity-amplitude recurrence. Extracting and controlling the vacuum and marginal gauge-action terms is required for an arbitrarily long quantum iteration. Large-field complements, mixed histories, observable estimates and continuum reconstruction remain parts of the existence argument.

References

[1] T. Bałaban, *Renormalization group approach to lattice gauge field theories. I*, Commun. Math. Phys. **109** (1987), 249–301. Original article copy. The local ordinary average and contour mean are taken from pp. 252–254; the composite-step arguments are developed here.

[2] *Toward a four-dimensional Yang–Mills construction*, research Notebooks 05–26 in this workspace. The classical minimum and gauge identities, flat quadratic locality, strict classical support families and the supported pre-gauge representative are the previously established inputs.

[3] E. Sussman and collaborators, Yang–Mills-LLM-experiment, deposited raw accounts at commit `e392ef45013a581664bab76b90867170b30fe10e`.